

Supplementary Material — The Changing Landscape of Medical Vision-Language Models: A Quantitative Survey of 1,217 arXiv Papers, 2024–2026

S1. arXiv queries (22). Each query was issued to the arXiv API over submissions dated 2024-01-01 to 2026-08-31; abs: restricts to the abstract field. Results were de-duplicated by arXiv identifier (latest version). Query 1 was split by year to stay below the API pagination limit. Repository: <https://github.com/fofifafo/medical-vlm-landscape-2026> (code, data, validation files).

Q1. abs:"vision-language model"

Q2. abs:"multimodal large language model"

Q3. abs:"visual instruction tuning"

Q4. abs:"efficient MLLM"

Q5. abs:"efficient vision-language"

Q6. abs:"token pruning" AND abs:multimodal

Q7. abs:"visual token" AND abs:compression

Q8. abs:"on-device" AND abs:"vision language"

Q9. abs:"mobile" AND abs:"vision-language model"

Q10. abs:"lightweight" AND abs:MLLM

Q11. abs:"KV cache" AND abs:multimodal

Q12. abs:quantization AND abs:MLLM

Q13. abs:"any resolution" AND abs:multimodal

Q14. abs:"high-resolution" AND abs:MLLM

Q15. abs:"document understanding" AND abs:multimodal

Q16. abs:"video" AND abs:MLLM

Q17. abs:"visual grounding" AND abs:"large language model"

Q18. abs:"multimodal hallucination"

Q19. abs:"multimodal" AND abs:"chain-of-thought"

Q20. abs:"retrieval-augmented" AND abs:multimodal

Q21. abs:"vision-language-action"

Q22. abs:"open-vocabulary" AND abs:segmentation

S2. VLM-relevance screen. A record was VLM-relevant when the title or abstract matched any of: multimodal large language model|MLLM|multi-modal LLM|multimodal LLM | vision[-]language model|VLM\b | visual instruction tuning (case-insensitive). 15,394 of 18,142 records.

S3. Medical filter. A record was medical when the following pattern matched the title, or matched the abstract at least twice (case-insensitive). 1,217 records.

medical|clinical|radiolog|patholog|patient|biomedical|chest x-?ray\b|CXR\b|CT scan\b|MRI\b\b|EHR\b|hospital|healthcare|health care|physician|nursing|surgery|surgical|oncolog|dermatolog|ophthalmolog|endoscop|histolog|whole[-]slide|ultrasound|fundus\b|ECG\b

S4. Taxonomy patterns (title + abstract, case-insensitive). Rows of Table I are marked; other branches were computed but omitted from Table I for space.

A. 임상 영역·모달리티

방사선 — X-ray/CXR: chest x-?ray\|bCXR\b|radiograph|x-?ray

방사선 — CT/MRI (3D): \bCT\b\|bMRI\b|volumetric|3D medical|computed tomography|magnetic resonance

병리 — 조직/WSI: patholog|histolog|whole[-]slide\|bWSI\b|biopsy

안과 — 안저/OCT: ophthalm|fundus|retina\|bOCT\b

피부과: dermatolog|skin lesion|skin disease

내시경·수술 영상: endoscop|surgical (video|scene)|laparoscop|surgery video|colonoscop

초음파: ultrasound|sonograph|echocardio

생체신호 (ECG/EEG): \bECG\b\|bEKG\b\|bEEG\b|electrocardio

치과·구강: dental|oral (cavity|health)|tooth|teeth

전자의무기록·문서: \bEHR\b|electronic health record|clinical note|discharge summary|medical record

일반/다중 모달리티: multi[-]?modal medical|across modalities|diverse medical imaging|multiple imaging modalit

B. 과제

판독문 생성: report generation|radiology report|generate (a |the)?report|report writing

의료 VQA: \bVQA\b|visual question answering|question[-]answering

진단·분류: diagnos|classif|disease (detection|prediction|identification)

병변 탐지·그라운딩: lesion detection|grounding|localiz|bounding box|abnormality detection

분할: segment

검색·RAG: retriev\|bRAG\b|retrieval[-]augmented

임상 의사결정·추론: clinical (decision|reasoning)|differential diagnosis|treatment (plan|recommend)|triage

의학 교육·시험: medical (exam|licens|education)|USMLE|board exam|medical student

환자 소통·상담: patient (communication|education|conversation)|health assistant|chatbot|consultation

수술 보조: surgical (assist|guidance|phase|skill)|intraoperative

C. 방법

범용 VLM 파인튜닝 (LLaVA 계): LLaVA|fine[-]tun\w* (a |the)?(general|open|pretrained)|adapt\w* (a |the)?general

도메인 사전학습 (CLIP 계): BiomedCLIP|MedCLIP|PubMedCLIP|contrastive|CLIP

명령어 튜닝·데이터 구축: instruction[-]tun|instruction data|instruction[-]following dataset|curated dataset|we (build|construct|introduce) (a|an)?(new |large[-]scale)?dataset

검색 증강: retrieval[-]augmented|bRAG\b|knowledge (base|graph) (augment|enhanc|inject)

에이전트·도구 사용: \bagent|tool[-]use|tool[-]calling|multi[-]agent

추론 기법 (CoT/RL/GRPO/DPO): chain[-]of[-]thought|reinforcement learning\bRL\b\bGRPO\b\bDPO\b\bRLHF\b|R1[-]style|reasoning (model|trace|chain)

추론 언급 (bare, 어휘): reasoning

경량화·효율: efficien|lightweight|quantiz|distill\bLoRA\b|parameter[-]efficient|on[-]device|token (prun|compress)

연합·프라이버시: federated|privacy[-]preserving|differential privacy|de[-]identif

D. 관심사 (열린 과제 재료)

환각 (명시적): hallucinat

신뢰성 어휘 (넓게): hallucinat|factual|faithful|reliab

벤치마크·평가: benchmark|evaluation (framework|protocol|suite)|we evaluate|systematic evaluation

안전·오용: safety|jailbreak|adversarial|misuse|harm

공정성·편향: fairness|demographic|disparit|equit\b\bias(ed)? (against|toward|across)|algorithmic bias|racial|gender bias

설명가능성: explainab|interpretab|attention map|saliency

데이터 부족·희소: data[-]scarce|low[-]resource|few[-]shot|limited (annotat|label|data)

임상 검증·실사용: clinical (validation|trial|deployment|practice)|real[-]world|prospective|reader study|radiologist(s)? (evaluat|rat|prefer)

규제·윤리: \bregulat(ory|ion|ions|ors?)\b\bFDA\b\bethic|accountab\b|liabilit

S5. Strict definitions

grounding_contribution: (propose|present|introduce)[^.]{0,200}(grounding|localiz|bounding box|lesion detection|abnormality detection)

clinical_validation_strict: reader study|prospective|clinical trial|(radiologist|clinician|physician)s? (evaluat|assess)|evaluat\w* by (w+)?(radiologist|clinician|physician)

S6. Efficiency-oriented set (144). Papers carrying at least one of the twelve keyword tags efficient-*vlm*, lightweight-*vlm*, small-*vlm*, on-device, mobile-*vlm*, low-power, token-pruning, token-merging, kv-cache, quantization, knowledge-distillation, peft-lora, assigned by the patterns in S2-style controlled vocabulary (*arxiv_scrape.py*). Full-text audit rubric: a quantity counts only when measured by the authors for their own system; related-work and cited values are excluded (*RUBRIC.md*).

S7. Classification validation. Judging model: Claude (Anthropic), interactive session, September 2026; deterministic reading of each abstract against the written criteria in *class_tags_guide.md*; one pass per paper; judgments stored in *class_judgments_claude.txt*. Efficiency audit judgments with supporting sentences: *audit.jsonl*. All files will be deposited with the final version.

S8. Classification-validation criteria (12 tags, verbatim from class_tags_guide.md). A tag is credited only when the paper performs the activity; the loose regex is shown for comparison.

분류 검증 12 개 태그 판정 안내

| 태그 | 판정 기준 | 규칙(정규식) |

| --- | --- | --- |

| 흉부 X 선 | 논문이 흉부 X 선(CXR/radiograph)을 다루는가 | `chest x-?ray | \bCXR\b | radiograph | x-?ray` |

| CT/MRI (3D) | 논문이 CT 또는 MRI 볼륨을 다루는가 | `\bCT\b | \bMRI\b | volumetric | 3D medical | computed tomography | magnetic resonance` |

| 병리/WSI | 병리 슬라이드·조직 영상을 다루는가 | `patholog | histolog | whole[- ]slide | \bWSI\b | biopsy` |

| 내시경·수술 영상 | 내시경 또는 수술 영상을 다루는가 | `endoscop | surgical (video | scene) | laparoscop | surgery video | colonoscop` |

| 판독문 생성 | 판독문(report) 생성이 과제인가 | `report generation | radiology report | generate (a | the)?report | report writing` |

| 그라운드·병변 위치 | 병변 위치 추정/그라운드·기여인가 (단어만 나오면 X) | `lesion detection | grounding | localiz | bounding box | abnormality detection` |

| 임상 의사결정·추론 | 임상 의사결정 지원/감별진단이 과제인가 | `clinical (decision | reasoning) | differential diagnosis | treatment (plan | recommend) | triage` |

| 추론 기법 (CoT/RL) | CoT·강화학습·GRPO/DPO 같은 구체 추론 기법을 쓰는가 | `chain[- ]of[- ]thought | reinforcement learning | \bRL\b | \bGRPO\b | \bDPO\b | \bRLHF\b | R1[- ]style | reasoning (model | trace | chain)` |

| 에이전트·도구 사용 | 에이전트 설계 또는 도구 호출을 쓰는가 | `\bagent | tool[- ]use | tool[- ]calling | multi[- ]agent` |

| 효율 기법 | 경량화·양자화·LoRA 등 효율 기법을 실제로 쓰는가 (형용사 'efficient'만이면 X) | `efficien | lightweight | quantiz | distill | \bLoRA\b | parameter[- ]efficient | on[- ]device | token (prun | compress)` |

| 환각 명시 | 환각(hallucination)을 명시적으로 다루는가 | `hallucin` |

| 임상 검증·실사용 | 임상의 평가·전향적 검증·실제 배포를 보고하는가 (희망 서술만이면 X) | `clinical (validation | trial | deployment | practice) | real[- ]world | prospective | reader study | radiologist(s)? (evaluat | rat | prefer)` |

S9. Sampling. Classification validation: 150 papers, 50 per year, drawn with seed 20260915 from the 1,217. Corpus-inclusion validation: 100 included papers (33/33/34 by year) plus 50 boundary exclusions (VLM-relevant records with exactly one clinical-term match in the abstract), seed 20260918; judged by the first author with disagreements adjudicated against the written criterion. Efficiency-audit verification: 30 papers stratified 15 with / 15 without a reported quantity, seed 20260914; all 180 judgments checked by the first author against the stored evidence sentences.

S10. Efficiency audit rubric (verbatim from RUBRIC.md).

효율 144 편 전문 판정 기준 (2026-09-14)

목적: 표 II 를 "초록 기준"에서 "본문 기준"으로 확장. 각 논문·각 항목에 대해 **저자가 자기 시스템을 측정해 보고한 값**이 본문(표·그림·문장)에 있는가만 판정한다.

항목 정의

| 코드 | 항목 | 인정 | 불인정 |

| --- | --- | --- | --- |

| LAT | 지연·추론 시간 | 자기 모델의 추론/학습 시간, ms/s per image, 속도 항상 배율(자기 실험) | 인용 논문 수치, "실시간" 같은 형용사만, 학습 epoch 수 |

| HW | 대상 하드웨어 | 실험에 쓴 GPU/CPU/엣지 기기 명시 | 관련연구의 기기 언급 |

| FLOP | 연산량 | 자기 모델 FLOPs/MACs 값 | 개념 언급만 |

| MEM | 메모리 | 자기 모델 GPU 메모리/VRAM/풋프린트 값 | 데이터셋 용량, 모델 파일 크기(비트 단위는 인정) |

| ENG | 에너지·전력 | 자기 측정 W, J, mWh, 전력/탄소 | "에너지 효율적"이라는 수식어, 배터리 언급만 |

| THR | 처리량 | 자기 모델 samples/s, QPS, fps | — |

| PAR | (보조) 파라미터 수 | 총/활성/학습 파라미터 값 | — |

판정 값

- `Y` 본문에 자기 측정값 있음 (근거 문장 발췌 필수)

- `A` 초록에만 있고 본문에 추가 없음 (초록 감사와 동일)

- `N` 없음

- `?` 판독 불가(PDF 변환 실패, 그림 안 숫자만) → 원문 페이지 확인 후 Y/N으로 확정

절 구분 규칙

- References 이후는 무시. Related work / Background 절의 문장은 타 논문 수치이므로 ****불인정****, 단 "we"/"our"가 주어이고 자기 모델 이름이 있으면 인정.

- 표 안의 수치는 표 캡션·열 이름으로 항목을 판단(예: "Time (s)" 열 → LAT).

산출물

- `audit.csv`: id, title, LAT, HW, FLOP, MEM, ENG, THR, PAR, any_Y, note

- `evidence.jsonl`: id 별 항목별 근거 문장(원문 발췌, 절 이름, 문서 내 위치)

- 검증용 표본 30 편: any_Y 여부로 층화 무작위 → 지유찬 판정과 κ

S11. Judging procedure. Each abstract (classification) or full text (efficiency audit) was read once by Claude Opus 5 in an interactive session in September 2026 with the criteria above as the instruction; the model returned one O/X per tag, or one Y/N per quantity with the supporting sentence. Outputs were stored verbatim (class_judgments_claude.txt, audit.jsonl) and not re-generated.

S12. Table II partition of the 144 efficiency-oriented papers: 28 report none of the six quantities; 65 name hardware and no other quantity; 11 report training time as their only quantity; 40 report at least one inference-relevant quantity (inference latency, FLOPs/MACs, memory, throughput). $28 + 65 + 11 + 40 = 144$. Of the 42 papers with any latency figure, 27 report inference latency and 15 training time only.